Learn money by playing it

A mobile-first financial-literacy game for Grade 11–12 students in underserved Hyderabad communities

1 The problem and our answer

Teenagers who skip a lecture on “compounding” will still play a game. The real barrier is **relevance and trust, not information**: a safe place to make a choice, see its consequence and try again. Minar is a walkable, hand-painted Hyderabad island. Four places each open a short game for one pillar: **Plan & Budget, Save & Manage, Understand Investing, Stay Safe**. Finishing a pillar lights a lamp on the Charminar; lighting all four opens the gate to the player’s goal (a fee, a course, a first phone).

We grade judgement, never profit. Every game scores decisions (did you check before you bid, did you keep money for a surprise) and builds one six-skill profile: planning, prioritising, saving, risk sense, patience, scam sense. Hard guardrails: we **never recommend investments**, every number is made up and labelled so, there are no real brands, no sign-in and no personal data.

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pillar games, all playable	languages: EN · हि · ఆ	personas, 2 avatars	sign-ins, no backend	automated tests, all passing



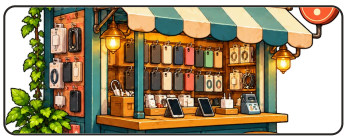
Plan & Budget
bangle market



Save & Manage
Irani café



Understand Investing
pearl shop



Stay Safe
recharge / UPI kiosk

2 What the player experiences

- **Home and story.** Named save slots, then a short skippable story told by the player’s grandmother: the fee letter that starts the goal. Players pick a **persona** (school, college or first-job earner) and an avatar; the persona sets starting money and goal.
- **Walk, tap, play.** Joystick or tap-to-walk on the island, pinch to zoom. No manual; works at 360 px on a low-end phone and on a laptop.
- **Two numbers on screen: Money and Happy.** Each lamp is worth a quarter of the gap to the goal, but its brightness depends on the player’s Happy Index (10–100, starts at 70): bright counts 100%, dim 75%, flickering 50%. Money earned by burning out is worth less.
- **Consequence, then retry.** Each game ends with a plain-words debrief: what happened, the idea, a one-line rule. A replay can only improve a pillar, so players improve choices, not luck: the same seed gives the same run.
- **Language-ready.** All copy lives in locale files (English, Hindi, Telugu). Every signal is icon plus label, and no state depends on colour alone.
- **Offline-first.** Static, installable web app; open it once on Wi-Fi and it works afterwards.
- **NGO-integrated.** A class code ties a cohort together; students can share an anonymous progress code. See section 7.

3 The four games

Pillar and game	What the player does	What it teaches, and how it is graded
1 Plan & Budget Festival Showdown	Treasurer of a colony festival: ₹50,000 on paper, but only ₹20,000 is in the bank. Pick and rank three goals, shop six jobs from four vendors (haggle, pay later, read the fine print, 10 actions), then live five money dates and a ten-futures stress test.	Cash flow, priorities, fixed vs variable cost, total cost, stress test, sales traps. Score out of 100 from goals, on-time bills, resilience and cost-smartness.
2 Save & Manage Pocket Panic	A six-minute boss fight where the boss is the month: split ₹600 into three jars, then a Hyderabad café throws temptation, small leaks, people and finally the rain at you. Chacha’s ledger shows where every rupee went.	Needs vs wants, small leaks, emergency fund, lending. Pressure bottles and a mood bar make over-saving and over-spending both visible.
3 Understand Investing Deal Board	A Monopoly-style ring of eight quarters. Meet one fictional company a quarter, run up to two checks (cash, customers, debt, the boss’s shares), then invest or pass and say why. Wildcards: hot tips, a fall, an emergency.	Research, scepticism, spread, calm, money habits (20 each); pass mark 60, never based on profit. Full-screen verdict: good, mixed, risky calls.
4 Stay Safe Case Files: Mohalla Detective	Scam triage: read fake offers, OTP and link messages, collect clues on a journey map, earn shield pieces, print a playbook.	Red flags and scam sense; content audit plus scripted end-to-end sweeps.

4 Plan & Budget in detail: Festival Showdown

This is the game we rebuilt last, after playtests told us the first version was too complex and did not feel like the project. Its design rules came from those playtests.

Explain where the choice is

Jargon was the main complaint (“what is a cushion?”), so “cushion” became **spare money for surprises**, and every action (haggle, pay later, fine print) carries a one-line plain explanation. A “Will your money last?” chart shows the festival fund after each date’s bills as bars: green is sure money, striped is promised money that may not come.

Show it, do not tabulate it

Each of the five money dates is a before-and-after picture: money in as a big green card, bills out as red cards, the fund as a bar. The “ten possible festivals” are **ten diyas to light**: lit means money held and goals met, dim means money held but a goal missed, out means money ran out. Tap a diya for its story.

Honest model

A cheap plan that never runs out of cash can still skip what people came for, so every future shows **goals met** as well as money held.

Learning from play

Four guesses (price rise, which sound costs less, can we pay Week 3, combo vs parts) are revealed against the truth; lessons fire in the player’s own numbers.

Same style everywhere

Painted characters shared with the hub, a layout that is one column on a phone and two columns on a laptop, and a full EN / HI / TE string table.

5 How it is built

Onboard persona, language › **Hub island** walk, tap, enter › **Game shell** registry, wrapper › **Seeded engine** pure functions › **Adapter** one result format › **Progress and Happy** lamps, skills › **Report and NGO view**

- **Stack:** Vite, React, TypeScript, Tailwind, Zustand, react-i18next, Vitest. Static hosting, no backend, so it scales to any cohort and costs nothing to run.
- **Engine and screen are separate.** Each game’s money logic is pure TypeScript over a seeded random generator, so a run is reproducible, testable and fair to replay. Screens only record decisions; the state is rebuilt from the decision log, so a reload resumes the exact screen.
- **A shell that treats every game the same.** A registry maps each pillar to a wrapper; a per-game adapter turns that game’s own result into one normalised result; one progress model turns results into lamps, skills and the Happy Index. Adding or swapping a game changes one folder, which is how we replaced two games late without touching the hub.
- **Tunable in two files:** the progress numbers and the Happy Index numbers each live in a single config file.

6 Results: what we measured

Layer	What we checked	Result
Whole app	Type-check, production build and the full test suite on the merged main branch	**620 tests in 33 files, all pass**; build is clean (19 MB, mostly painted art)
Plan & Budget engine	Our TypeScript engine compared with the Python reference prototype from the spec: 70 plans across all ten futures, plus the spec’s reference numbers	Exact match. Careful plan: **98/100, survives 9 of 10** futures. Thin plan with one late payment: **45/100, survives 3 of 10**
Plan & Budget languages	517 strings in EN, HI and TE; a test checks every key and every {{placeholder}} exists in all three	Passes. No English left on the report or date screens in Telugu
Balance (Deal Board)	Bots over 1,000 seeds: a bot that spreads money equally and never researches	Scores 35–55, below the pass mark of 60, so skill beats luck
Shell and progress	Result adapters, progress and Happy rules, host flow, saves, NGO analytics	Adapters 75, progress 53, Happy 42, host 42, NGO 13 tests
Pocket Panic, Case Files	Month engine and strings (29 tests); content validator (7 cases, 0 errors), scripted play-throughs, offline and responsive checks	Pass
Guardrails	“Made-up numbers. Real life is different.” shown on every simulation screen; no personal data stored	Present on all four games

7 The NGO view: a facilitator can see learning without seeing a student

- **Class code and an anonymous progress code.** A student can share a code that holds no name and nothing they typed except the class code; it survives being pasted inside a chat message and rejects tampered codes.
- **Facilitator screens:** overview, skills, the lesson to teach next, wellbeing, a list of students who still have places to clear, a rollout checklist (open the game once on Wi-Fi, write the class code on the board) and a printable report.
- **Privacy by design:** groups smaller than five are never broken out; CSV export is one anonymous line per student; a built-in demo class lets a facilitator see the view before any real data exists.

8 How Minar meets the seven success measures

Measure	How it is delivered	Honest status
Simple and intuitive	Walk, tap, play; one decision per screen; inline explanation next to every unusual choice; plain words, icon plus label	Playtested internally and revised repeatedly (the Plan & Budget game was rebuilt twice from feedback); no students yet
Relatable	A Hyderabad bazaar, ₹ amounts, chai, UPI, recharge scams, a colony festival, an Irani café	Done
Persona-led	School, college and first-job personas set starting money and goal; the Happy Index reacts to how the player plays	**Partial.** Start money and goals scale by persona; per-game difficulty scaling is the next step
Language-ready	English, Hindi, Telugu from one string table per game, plus ₹ formatting by language	**Machine-drafted Hindi and Telugu still need a native-speaker review.** Part of the Stay Safe game still shows English where Telugu is missing
Game-based evaluation	Every game gives a judgement score and concept evidence, not a quiz; guesses are revealed against the truth; replay shows improvement	Done in all four games
NGO-integrated	Class code, anonymous progress code, facilitator dashboard, rollout checklist, anonymous CSV	Built and tested on a demo class
Works on cheap phones, offline	Static app, 360 px layouts, lite mode, installable web app with a service worker	Works offline; the first download is large (19 MB, mostly art) and should be slimmed for patchy networks

9 Limits we are stating plainly

- **No pilot with real students yet.** Everything above is from automated tests, bots and internal playtests. The first job after the hackathon is a pilot with the NGO partner and a before-and-after question set on the same cohort.
- **Cross-game checks still to run.** The balance sweep (bots over many seeds) and the pre/post learning questions were done per game, not yet across the merged app.
- **Human review pending.** Hindi and Telugu wording, and the lesson texts, need a native speaker and a teacher's read before classroom use.
- **Two offline workers.** The shell and the Stay Safe game each register a service worker; we plan to fold them into one and precache only what a first session needs.

10 What is next

Persona depth
Scale amounts and difficulty inside each game by persona, and show one visible adaptive behaviour.

Lighter and safer offline
One service worker, a smaller first download, and a real low-end-phone test.

Evidence
A short pre/post set, a teacher review, native Hindi and Telugu proofreading, then a pilot with the NGO partner.

In one sentence: Minar lets a teenager make a money decision, feel the consequence, retry, and walk away with a skill profile a facilitator can read without ever learning who the student is, and it does so on a low-end phone, offline, in three languages, with no backend.

Where to find it

All code is in the team repository on **main**. **README.md** has the install and run steps (`npm install`, `npm run dev`, `npm test`, `npm run build`). Each game folder has its own tests; Festival Showdown has a checking guide with links that jump to any screen. The demo video shows one full loop: story, island, a game, the lamp lighting, and the facilitator view.